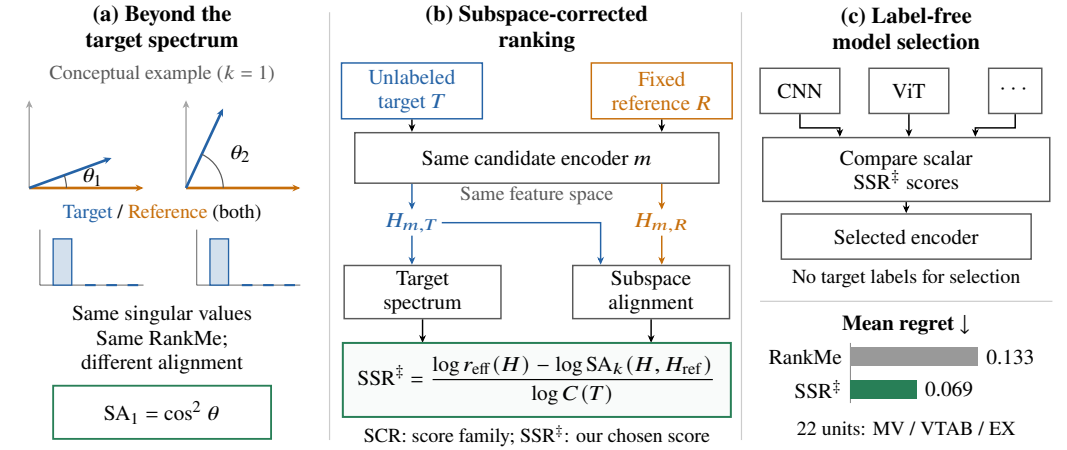
(a) Beyond the target spectrum
Conceptual example ($k = 1$)
θ_1
θ_2
Target / Reference (both)
Same singular values
Same RankMe;
different alignment
$\mathrm{SA}_1 = \cos^2\theta$
(b) Subspace-corrected ranking
Unlabeled target T
Fixed reference R
Same candidate encoder m
Same feature space
$H_{m,T}$
$H_{m,R}$
Target spectrum
Subspace alignment
$\mathrm{SSR}^{\ddagger} = \frac{\log r_{\mathrm{eff}}(H) - \log \mathrm{SA}_k(H, H_{\mathrm{ref}})}{\log C(T)}$
SCR: score family; $\mathrm{SSR}^{\ddagger}$: our chosen score
(c) Label-free model selection
CNN
ViT
· · ·
Compare scalar $\mathrm{SSR}^{\ddagger}$ scores
Selected encoder
No target labels for selection
Mean regret ↓
RankMe
0.133
$\mathrm{SSR}^{\ddagger}$
0.069
22 units: MV / VTAB / EX